ALPER YILDIRIM

AI Researcher — Mechanistic Interpretability, Geometric Learning & Architecture Design

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Research Profile

AI researcher studying how neural networks represent and process information through causal analysis of features, circuits, phase, geometry, and topology. My work spans mechanistic interpretability, geometric deep learning, and topological deep learning; where useful, I translate these findings into architecture design and inductive biases. I investigate these questions across language, vision, learning dynamics, time series, and optical computing.

Publications & Selected Research

Language as a Wave Phenomenon: Semantic Phase Locking and Interference in Neural Networks | *ICML 2026 Main Track — accepted and presented — A. Yildirim, I. Yucedag*

Introduces PRISM, a complex-valued FFT sequence architecture with phase-preserving primitives; variants outperform a Transformer baseline with fewer parameters.

The Importance of Phase in Neural Representations: An Internal Oppenheim–Lim Test of Image Classifiers | *Sole author — preprint*

Layer-resolved causal phase/magnitude transplants show that class identity follows phase/sign while image-specific magnitude is often dispensable.

The Geometric Inductive Bias of Grokking: Bypassing Phase Transitions via Architectural Topology | *Sole author — under review at a top-tier machine learning conference*

Spherical L2 residual topology removes memorization in modular arithmetic but fails on non-commutative S5, linking generalization gains to geometry–task alignment.

Superposition Is Not Necessary: Mechanistic Interpretability of Transformer Representations for Time Series | *Sole author — under review*

Sparse autoencoders and causal probing in PatchTST examine feature structure, interpretability faithfulness, and linear representations in forecasting.

HAMON: Passive Optical Sequence Mixing for Long-Horizon Forecasting | *Sole author — 2 pending Turkish patent applications*

Trainable phase masks perform passive diffractive sequence mixing, connecting learned phase codes to optical hardware through differentiable simulation and causal controls.

CRT Structure in Transformer Addition | *Open research artifact*

Causal interventions decompose Pythia-6.9B arithmetic into residue and magnitude channels, revealing Chinese Remainder Theorem structure.

Research & Industry Experience

Research Collaborations — current

Atticus Geiger (Goodfire) and Patrick Lutz (Boston University)

Causal analysis of AI models and the algorithms they discover.

AI Engineer & Lead Researcher — isTechSoft

Jun 2024 – Jul 2025

Secured TUBITAK 1505 funding and led execution from diagnostics research to production. Fine-tuned domain LLMs with RAG, SFT, and GRPO; built evaluation pipelines and deployed an LLM, embedding model, and reranker on a single 16 GB GPU.

Selected Projects & Patents

- Optical forecasting system — two pending Turkish patent applications.
- Neural network implemented from scratch in C++.
- U-Net earthquake rupture detection from drone footage (>95% original evaluation accuracy).
- EEG-based Pathologic Echo Hypothesis preprint.

Technical Skills

Research: causal interventions, activation patching, SAEs, representation geometry, Fourier/phase analysis, circuit analysis, grokking, geometric and topological inductive biases.

ML: PyTorch, Hugging Face, TransformerLens-style workflows, vLLM, LoRA/QLoRA, RAG, SFT/GRPO, TorchOptics; Transformers, ViT, ResNet, Mamba, U-Net, TFT.

Engineering: Python, C++, C#, SQL, Docker, FastAPI, Flask, Git, GPU optimization, production deployment.

Education

B.Sc. Computer Engineering

Duzce University — 2026

First-author ICML main-track paper completed during undergraduate study, followed by independent sole-author research in mechanistic interpretability, geometric learning, and architecture design.